

# Lecture 6: Money in Utility Models

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## Model I. Classical monetary model

- 1 assume perfect competition & fully flexible prices
- 2 limited role for money, real and nominal dichotomy:

$$N, C, Y, K, \frac{R}{P}, \frac{M}{P}$$

pinned down without monetary policy. Monetary policy only affects nominal variables.

## Household's problem

$$\max_{\{C_t, N_t, B_t\}} E_0 \sum_{t=0}^{\infty} \beta^t U(C_t, N_t)$$

s.t.

$$P_t C_t + Q_t B_t = B_{t-1} + W_t N_t + T_t$$

where

$B_t$  : one-period, nominally riskless discount bonds that mature in one period; bought in  $t$ , and pay one unit of money at  $t + 1$ .

$Q_t$  : price of bonds; pay  $Q_t$  units today and get 1 unit next period,  
or, pay 1 unit today and get  $\frac{1}{Q_t}$  units, i.e. gross return next period.

$T_t$  : lump-sum taxes (if  $T_t < 0$ ) or dividends (if  $T_t > 0$ ).

Lagrangian:

$$\mathcal{L} = E_0 \sum_{t=0}^{\infty} \beta^t \left[ U(C_t, N_t) + \lambda_t (B_{t-1} + W_t N_t - T_t - P_t C_t - Q_t B_t) \right].$$

FOCs:

$$\frac{\partial \mathcal{L}}{\partial C_t} = 0 \quad \Rightarrow \quad U_{C_t} - \lambda_t P_t = 0 \quad (1)$$

$$\frac{\partial \mathcal{L}}{\partial N_t} = 0 \quad \Rightarrow \quad U_{N_t} + \lambda_t W_t = 0 \quad (2)$$

$$\frac{\partial \mathcal{L}}{\partial B_t} = 0 \Rightarrow -\beta^t \lambda_t Q_t + \beta^{t+1} E_t[\lambda_{t+1}] = 0 \Rightarrow Q_t = \beta E_t \left[ \frac{\lambda_{t+1}}{\lambda_t} \right] \quad (3)$$

$$\frac{\partial \mathcal{L}}{\partial \lambda_t} = 0 \Rightarrow P_t C_t + Q_t B_t = B_{t-1} + W_t N_t + T_t. \quad (5)$$

From the first two FOCs:

$$-\frac{U_{N_t}}{U_{C_t}} = \frac{W_t}{P_t}.$$

From the bond FOC:

$$Q_t = \beta E_t \left[ \frac{U_{C,t+1}}{U_{C,t}} \frac{P_t}{P_{t+1}} \right].$$

Suppose

$$U(C_t, N_t) = \frac{C_t^{1-\sigma}}{1-\sigma} - \frac{N_t^{1+\varphi}}{1+\varphi}.$$

Then

$$U_{C_t} = C_t^{-\sigma}, \quad U_{N_t} = -N_t^\varphi.$$

So

$$\frac{W_t}{P_t} = C_t^\sigma N_t^\varphi.$$

Hence

$$N_t = \left( \frac{W_t/P_t}{C_t^\sigma} \right)^{1/\varphi}.$$

$$Q_t = \beta E_t \left[ \left( \frac{C_{t+1}}{C_t} \right)^{-\sigma} \frac{P_t}{P_{t+1}} \right].$$

$$\frac{W_t}{P_t} = C_t^\sigma N_t^\varphi \Rightarrow N_t = \left( \frac{W_t/P_t}{C_t^\sigma} \right)^{1/\varphi} \Rightarrow \log N_t = \frac{1}{\varphi} \log \left( \frac{W_t}{P_t} \right) - \frac{\sigma}{\varphi} \log C_t \Rightarrow \frac{\partial \log N_t}{\partial \log (W_t/P_t)} = \frac{1}{\varphi}$$

$\frac{1}{\varphi}$ : Frisch elasticity (how labor supply responds to real wage),  $\varphi \uparrow \Rightarrow$  elasticity  $\downarrow$ .

**Firm's problem**

$$\max_{Y_t, N_t} \Pi_t = P_t Y_t - W_t N_t$$

s.t.

$$Y_t = A_t N_t^{1-\alpha}.$$

FOC:

$$\frac{W_t}{P_t} = (1-\alpha) A_t N_t^{-\alpha} \quad : \text{both are real terms.} \quad (4)$$

**Market clearing**

$$\text{Goods market clearing: } Y_t = C_t$$

$$\text{Labor market clearing: } N_t^{(d)} = N_t^{(s)} \text{ (implicitly holds)}$$

$$\text{Bond market clearing: Hold by Walras Law } B_t = 0$$

Then we have the system

$$\left\{ \begin{array}{ll} Y_t = C_t, & \text{goods market clearing} \\ T_t = \alpha P_t Y_t, & \text{dividend} \\ Y_t = A_t N_t^{1-\alpha}, & \text{production function} \\ \frac{W_t}{P_t} = C_t^\sigma N_t^\varphi, & \text{labor-leisure condition} \\ \frac{W_t}{P_t} = (1-\alpha) A_t N_t^{-\alpha}, & \text{real wage = MPL} \\ P_t C_t + Q_t B_t = B_{t-1} + W_t N_t + T_t, & \text{budget constraint} \\ Q_t = \beta E_t \left[ \left( \frac{C_{t+1}}{C_t} \right)^{-\sigma} \frac{P_t}{P_{t+1}} \right], & \text{bond Euler equation} \\ \log A_t = (1-\rho_a) \log \bar{A} + \rho_a \log A_{t-1} + \varepsilon_t, & \text{productivity process} \end{array} \right.$$

No monetary policy involved. Denote the real wage as  $W_t^r = \frac{W_t}{P_t}$ .

$$\Rightarrow \quad \text{Real equilibrium (in units of goods):} \quad \left\{ \begin{array}{ll} W_t^r = C_t^\sigma N_t^\varphi & (1) \\ Y_t = A_t N_t^{1-\alpha} & (2) \\ W_t^r = (1-\alpha) A_t N_t^{-\alpha} & (3) \\ Y_t = C_t & (4) \end{array} \right.$$

Four unknowns:  $W_t^r, C_t, N_t, Y_t$ . Four equations.

**At steady state** Assume

$$\bar{A} = 1.$$

From (2), (3), and  $Y = C$ :

$$(1-\alpha) \bar{N}^{-\alpha} = \bar{C}^\sigma \bar{N}^\varphi.$$

Since

$$\bar{Y} = \bar{N}^{1-\alpha}, \quad \bar{C} = \bar{Y},$$

we have

$$(1-\alpha) \bar{N}^{-\alpha} = (\bar{N}^{1-\alpha})^\sigma \bar{N}^\varphi = \bar{N}^{\sigma(1-\alpha)+\varphi}.$$

Hence

$$1-\alpha = \bar{N}^{\sigma(1-\alpha)+\varphi+\alpha},$$

so

$$\bar{N} = (1-\alpha)^{\frac{1}{\sigma(1-\alpha)+\varphi+\alpha}}.$$

Thus

$$\bar{Y} = \bar{N}^{1-\alpha} = (1-\alpha)^{\frac{1-\alpha}{\sigma(1-\alpha)+\varphi+\alpha}}, \quad \bar{C} = \bar{Y}.$$

and

$$\bar{W}^r = \bar{C}^\sigma \bar{N}^\varphi = (1-\alpha) \bar{N}^{-\alpha}.$$

## Log-linearization

$$\begin{aligned}\hat{w}_t^r &= \sigma \hat{c}_t + \varphi \hat{n}_t \\ \hat{y}_t &= \hat{a}_t + (1 - \alpha) \hat{n}_t \\ \hat{w}_t^r &= \hat{a}_t - \alpha \hat{n}_t \\ \hat{y}_t &= \hat{c}_t.\end{aligned}$$

Only one shock  $\hat{a}_t$ , no predetermined variable. Assume

$$\hat{y}_t = \psi_{ya} \hat{a}_t, \quad \hat{n}_t = \psi_{na} \hat{a}_t, \quad \hat{w}_t^r = \psi_{wa} \hat{a}_t$$

Then

$$\sigma \hat{a}_t + \sigma(1 - \alpha) \hat{n}_t + \varphi \hat{n}_t = \hat{a}_t - \alpha \hat{n}_t$$

$$\Rightarrow \quad \hat{c}_t = \hat{y}_t = \frac{1 + \varphi}{\sigma(1 - \alpha) + \varphi + \alpha} \hat{a}_t \quad \text{coefficient} > 0$$

$$\hat{w}_t^r = \frac{\varphi + \sigma}{\sigma(1 - \alpha) + \varphi + \alpha} \hat{a}_t \quad \text{coefficient} > 0$$

$$\hat{n}_t = \frac{1 - \sigma}{\sigma(1 - \alpha) + \varphi + \alpha} \hat{a}_t \quad \text{coefficient is ambiguous, depends on } \sigma$$

Income effect dominates if  $\sigma > 1$ :  $\frac{\partial \hat{n}_t}{\partial \hat{a}_t} < 0$ .

Substitution effect dominates if  $\sigma < 1$ :  $\frac{\partial \hat{n}_t}{\partial \hat{a}_t} > 0$ .

**Real interest rate** Bond Euler:

$$Q_t = \beta E_t \left[ \left( \frac{C_{t+1}}{C_t} \right)^{-\sigma} \frac{1}{\Pi_{t+1}} \right]. \quad (*)$$

$\left( \frac{C_{t+1}}{C_t} \right)^{-\sigma}$  : real stochastic discount factor,  $\frac{1}{\Pi_{t+1}}$  : converts  $t+1$  nominal payoff into real terms.

Let

$$Q_t = e^{-i_t}, \quad \frac{P_{t+1}}{P_t} = \Pi_{t+1} \text{ inflation}, \quad \pi_{t+1} = \log \Pi_{t+1} = \log P_{t+1} - \log P_t, \quad \beta = e^{-\rho}$$

Define real interest rate:  $r_t = i_t - E_t \pi_{t+1}$

$i_t = -\log Q_t$  log nominal interest rate

$\rho = -\log \beta$  rate of time preference

Define new deviation:  $\hat{i}_t = i_t - \bar{i} \quad \hat{r}_t = r_t - \bar{r} \quad \bar{\pi} = \log \bar{P} - \log \bar{P} = 0$

$$\left\{ \begin{aligned} \left( \frac{C_{t+1}}{C_t} \right)^{-\sigma} &= e^{-\sigma \log C_{t+1}} / e^{-\sigma \log C_t} = e^{-\sigma(\log C_{t+1} - \log C_t)}, \\ \frac{1}{\Pi_{t+1}} &= \frac{P_t}{P_{t+1}} = e^{\log P_t - \log P_{t+1}} = e^{-\pi_{t+1}}. \end{aligned} \right.$$

$$\Rightarrow e^{-i_t} = e^{-\rho} E_t \left[ e^{-\sigma(\hat{c}_{t+1} - \hat{c}_t) - \pi_{t+1}} \right] \quad (**)$$

$$-i_t = -\rho + E_t[-\sigma(\hat{c}_{t+1} - \hat{c}_t) - \pi_{t+1}] \quad (1)$$

$$\left. \begin{aligned} -\bar{i} &= -\rho + E_t[-\sigma(\log \bar{C} - \log \bar{C}) - \bar{\pi}] \\ \bar{\pi} &= \log \bar{P} - \log \bar{P} = 0 \end{aligned} \right\} \Rightarrow \bar{i} = \rho \quad (2)$$

$$(1) + (2) \Rightarrow -\hat{i}_t + \bar{i} = E_t[-\sigma(\log C_{t+1} - \log C_t) - \pi_{t+1}].$$

$$\begin{cases} \hat{c}_{t+1} = \log C_{t+1} - \log \bar{C}, \\ \hat{c}_t = \log C_t - \log \bar{C} \end{cases} \Rightarrow \hat{c}_{t+1} - \hat{c}_t = \log C_{t+1} - \log C_t.$$

$$\Rightarrow -\hat{i}_t = E_t[-\sigma(\hat{c}_{t+1} - \hat{c}_t) - \pi_{t+1}],$$

$$\Rightarrow -\hat{i}_t = -\sigma E_t \Delta \hat{c}_{t+1} - E_t[\pi_{t+1}],$$

$$\Rightarrow \sigma E_t \Delta \hat{c}_{t+1} = \hat{i}_t - E_t[\pi_{t+1}], \quad (***)$$

$$\hat{\pi}_{t+1} = \pi_{t+1} - \bar{\pi} = \pi_{t+1},$$

$$\Rightarrow \hat{i}_t - E_t[\hat{\pi}_{t+1}] = \sigma E_t \Delta \hat{c}_{t+1}.$$

$$\text{Since } r_t = i_t - E_t[\pi_{t+1}], \quad (3)$$

$$\bar{r} = \bar{i} - \bar{\pi} = \bar{i}, \quad (4)$$

$$(3) - (4) \Rightarrow \hat{r}_t = \hat{i}_t - E_t[\hat{\pi}_{t+1}]$$

$$\text{Plug in } (***) \Rightarrow \hat{r}_t = \sigma E_t \Delta \hat{c}_{t+1}$$

$$\hat{c}_{t+1} = \hat{y}_{t+1} = \psi_{ya} \hat{a}_{t+1},$$

$$\Rightarrow \hat{r}_t = \sigma \psi_{ya} E_t[\Delta \hat{a}_{t+1}]$$

$$\text{If the shock is transitory / mean reverting} \Rightarrow E_t[\Delta \hat{a}_{t+1}] < 0 \Rightarrow r_t < 0.$$

A positive transitory technology shock causes the current real interest rate to fall below its steady-state level.

Intuition:  $a_t \uparrow$  gives a favorable temporary income shock  $\Rightarrow$  households smooth consumption by saving  $\Rightarrow$  bond demand rises  $B_t^d \uparrow \Rightarrow$  bond price rises  $Q_t \uparrow \Rightarrow$  real interest rate falls  $r_t \downarrow$ .

## Dichotomy

- $r_t$  pinned down by real side, money growth does not affect  $r_t$ .
- No policy is the best since only real variables enter utility.
- Does not match empirics: data say real variable move after monetary shocks and nominal variables respond slugg.

## Model II. Money in utility model

- Make nominal terms affect real terms.
- Explain why agents hold money which is inferior to bond. (Money has no return but affects utility.)

### Household's Problem

$$\max_{\{C_t, M_t, N_t\}_{t=0}^{\infty}} E_0 \sum_{t=0}^{\infty} \beta^t U(C_t, M_t, N_t)$$

$$\text{s.t. } P_t C_t + Q_t B_t + M_t \leq B_{t-1} + M_{t-1} + W_t N_t + T_t$$

Let  $\mathbb{M}_t = \frac{M_t}{P_t}$ , the real money balance/supply.

Financial wealth before decisions(predetermined variable):

$$A_t = B_{t-1} + M_{t-1}$$

$$\Lambda_{t,T} = Q_t \cdot Q_{t+1} \cdots Q_T$$

B.C.:

$$P_t C_t + Q_t A_{t+1} + (1 - Q_t) M_t = A_t + W_t N_t + T_t$$

Notes 1:

$i_t$  : continuously compounded nominal rate

$Q_t$  : price today  $Q_t$ , payoff tomorrow 1

Nominal interest rate:

$$1 - Q_t = 1 - e^{-i_t} \approx i_t$$

$1 - Q_t$  : the opportunity cost of holding one's financial wealth in money rather than in interest-bearing bonds; a high  $i_t$  thus means contractionary monetary policy.

Notes:

$\Lambda_{t,T}$  : state-contingent discount factor from  $t$  to  $T$

$\frac{A_T}{P_T}$  : terminal real wealth

No-Ponzi scheme:

$$\lim_{T \rightarrow \infty} E_t \left[ \Lambda_{t,T} \frac{A_T}{P_T} \right] \geq 0, \quad \forall t$$

Notes:

cannot end up with infinitely bad wealth

(borrow a lot  $\Rightarrow$  roll-over)

TVC:

$$\lim_{T \rightarrow \infty} E_t \left[ \Lambda_{t,T} \frac{A_T}{P_T} \right] = 0$$

Note:

no resources on table when world ends

All financial assets  $A_t$  yield gross nominal return  $Q_t^{-1} = e^{i_t}$ .

Lagrangian:

$$\mathcal{L} = E_0 \sum_{t=0}^{\infty} \beta^t \left[ U(C_t, \mathbb{M}_t, N_t) + \lambda_t (A_t + W_t N_t + T_t - P_t C_t - Q_t A_{t+1} - (1 - Q_t) M_t) \right].$$

FOCs:

$$\frac{\partial \mathcal{L}}{\partial M_t} = 0 \Rightarrow U_{\mathbb{M}_t} \frac{1}{P_t} - \lambda_t (1 - Q_t) = 0$$

$$\frac{\partial \mathcal{L}}{\partial C_t} = 0 \Rightarrow U_{C,t} = \lambda_t P_t$$

$$\frac{\partial \mathcal{L}}{\partial N_t} = 0 \Rightarrow -\frac{U_{N,t}}{U_{C,t}} = \frac{W_t}{P_t}.$$

Hence

$$\frac{U_{\mathbb{M}_t}}{U_{C_t}} = 1 - Q_t = 1 - e^{-i_t}. \quad (5)$$

### Separable utility case(still dichotomy)

Assume

$$U(C_t, \mathbb{M}_t, N_t) = \frac{C_t^{1-\sigma}}{1-\sigma} + \frac{(\mathbb{M}_t)^{1-\nu}}{1-\nu} - \frac{N_t^{1+\varphi}}{1+\varphi}.$$

Neither  $U_{C_t}$  nor  $U_{N_t}$  depends on  $\mathbb{M}_t \Rightarrow$  real equilibrium not affected(same as before).

But we can derive money demand:

$$U_{\mathbb{M}_t} = (\mathbb{M}_t)^{-\nu}, \quad U_{C_t} = C_t^{-\sigma}.$$

$$\frac{U_{\mathbb{M},t}}{U_{C,t}} = 1 - e^{-i_t} \Rightarrow \frac{\mathbb{M}_t^{-\nu}}{C_t^{-\sigma}} = 1 - e^{-i_t}$$

$$\Rightarrow \mathbb{M}_t = C_t^{\sigma/\nu} [1 - e^{-i_t}]^{-1/\nu}$$

$$\Rightarrow \log M_t - \log P_t = \frac{\sigma}{\nu} \log C_t - \frac{1}{\nu} \log(1 - e^{-i_t})$$

$$\hat{\mathbb{M}}_t \approx \frac{\sigma}{\nu} \hat{c}_t - \frac{1}{\nu (e^{\bar{i}} - 1)} \hat{i}_t.$$

math notes:

$$\text{Let } f(i) = \log(1 - e^{-i}),$$

$$f'(i) = \frac{e^{-i}}{1 - e^{-i}} = \frac{1}{e^i - 1},$$

$$f''(i) = -\frac{e^i}{(e^i - 1)^2},$$

$$\log(1 - e^{-i_t}) \approx \log(1 - e^{-\bar{i}}) + \frac{1}{e^{\bar{i}} - 1} (i_t - \bar{i}).$$

$\log(1 - e^{-\bar{i}})$  is constant, so it cancels out in derivation.

If  $\sigma = \nu$ , define

$$\eta \equiv \frac{1}{\nu (e^{\bar{i}} - 1)},$$

then approximately

$$\hat{m}_t - \hat{p}_t = \hat{y}_t - \eta \hat{i}_t.$$

$\eta$ : semi-elasticity of real money demand with respect to the nominal interest rate.

$\hat{m}_t - \hat{p}_t$ : real money balances are determined by  $\hat{y}_t$  and  $\hat{i}_t$ .

$\hat{y}_t$ : determined by the real-side equilibrium, so monetary policy does not affect real activity in the separable case.

## Inseparable utility case

Utility:

$$U(C_t, \mathbb{M}_t, N_t) = \frac{X_t^{1-\sigma}}{1-\sigma} - \frac{N_t^{1+\varphi}}{1+\varphi}.$$

$$\text{where, } X_t = \begin{cases} [(1-\theta)C_t^{1-\nu} + \theta\mathbb{M}_t^{1-\nu}]^{\frac{1}{1-\nu}}, & \nu \neq 1, \\ C_t^{1-\theta}\mathbb{M}_t^\theta, & \nu = 1. \end{cases}$$

Elasticity of substitution between  $C_t$  and  $\mathbb{M}_t$  is  $\frac{1}{\nu}$

$\theta \in (0, 1)$  weight on money inside the CES aggregator.

Derivatives when  $\nu \neq 1$

For  $\nu \neq 1$ ,

$$X_t = [(1-\theta)C_t^{1-\nu} + \theta\mathbb{M}_t^{1-\nu}]^{\frac{1}{1-\nu}}.$$

Then

$$U_{C,t} = X_t^{-\sigma} \cdot \frac{1}{1-\nu} \cdot X_t^\nu \cdot (1-\theta)(1-\nu)C_t^{-\nu}$$

so

$$U_{C,t} = (1-\theta)X_t^{\nu-\sigma}C_t^{-\nu}.$$

Similarly,

$$U_{\mathbb{M},t} = X_t^{-\sigma} \cdot \frac{1}{1-\nu} \cdot X_t^\nu \cdot \theta(1-\nu)\mathbb{M}_t^{-\nu}$$

so

$$U_{\mathbb{M},t} = \theta X_t^{\nu-\sigma}\mathbb{M}_t^{-\nu}.$$

Also,

$$U_{N,t} = -N_t^\varphi.$$

**Marginal rate of substitution and elasticity** Define

$$X_C = (1 - \theta)X^\nu C^{-\nu}, \quad X_M = \theta X^\nu M^{-\nu}.$$

Then

$$MRS_{C,M} = \frac{U_C}{U_M} = \frac{X_C}{X_M} = \frac{1 - \theta}{\theta} \left( \frac{M}{C} \right)^\nu.$$

Hence

$$\varepsilon = \frac{d \ln(M/C)}{d \ln MRS_{C,M}} = \frac{1}{\nu}.$$

Thus,

$$\nu > 1 \Rightarrow \varepsilon < 1 \quad (\text{complements}),$$

$$\nu < 1 \Rightarrow \varepsilon > 1 \quad (\text{substitutes}).$$

**FOCs** Labor supply condition:

$$-\frac{U_{N,t}}{U_{C,t}} = \frac{W_t}{P_t}.$$

Substituting the derivatives,

$$\begin{cases} \frac{W_t}{P_t} = \frac{1}{1 - \theta} N_t^\varphi X_t^{\sigma - \nu} C_t^\nu, \\ Q_t = \beta E_t \left[ \frac{U_{C,t+1}}{U_{C,t}} \frac{P_t}{P_{t+1}} \right] = \beta E_t \left[ \left( \frac{C_{t+1}}{C_t} \right)^{-\nu} \left( \frac{X_{t+1}}{X_t} \right)^{\nu - \sigma} \frac{P_t}{P_{t+1}} \right]. \end{cases}$$

$X_t$  depends on  $M_t$ ,  
thus money is non-neutral.

Money FOC:

$$\frac{U_{M,t}}{U_{C,t}} = 1 - e^{-i_t}.$$

So

$$\frac{\theta}{1 - \theta} C_t^\nu M_t^{-\nu} = 1 - e^{-i_t}.$$

Solving for real money balances,

$$M_t = C_t \left( 1 - e^{-i_t} \right)^{-1/\nu} \left( \frac{\theta}{1 - \theta} \right)^{1/\nu}.$$

When  $\nu = \sigma$  and we drop  $\theta$ , the utility function reduces to the separable case, so the dichotomy holds.

When  $\nu \neq \sigma$ , we have the general (non-separable) case.

- 1) Labor supply and Euler equation are affected by the level of real balances through  $X_t$ ,
- 2) Real balance depends on nominal interest rate.

Non-neutral monetary policy:      Monetary policy  $\Rightarrow$  nominal rate  $i_t$   
 $\Rightarrow$  real balance  $M \Rightarrow$  labor supply and output.

Log linearize:       $M_t = C_t (1 - e^{-i_t})^{-1/\nu} \left( \frac{\theta}{1 - \theta} \right)^{1/\nu} \Rightarrow \hat{m}_t - \hat{p}_t = \hat{c}_t - \eta \hat{i}_t \quad \dots (*)$

$$\frac{W_t}{P_t} = \frac{1}{1-\theta} N_t^\varphi X_t^{\sigma-\nu} C_t^\nu \Rightarrow \hat{w}_t - \hat{p}_t = \sigma \hat{c}_t + \varphi \hat{n}_t + (\nu - \sigma)(\hat{c}_t - \hat{x}_t) \quad \dots (**)$$

$$X_t^{1-\nu} = (1-\theta)C_t^{1-\nu} + \theta \bar{\mathbb{M}}^{1-\nu}$$

$$\text{S.S.: } (1-\theta)\bar{C}^{1-\nu} = \bar{X}^{1-\nu} - \theta\bar{\mathbb{M}}^{1-\nu}$$

$$\Rightarrow \bar{X}^{1-\nu} \hat{x}_t = (1-\theta)\bar{C}^{1-\nu} \hat{c}_t + \theta\bar{\mathbb{M}}^{1-\nu}(\hat{m}_t - \hat{p}_t).$$

$$\text{Using } (1-\theta)\bar{C}^{1-\nu} = \bar{X}^{1-\nu} - \theta\bar{\mathbb{M}}^{1-\nu},$$

$$\bar{X}^{1-\nu} \hat{x}_t = [\bar{X}^{1-\nu} - \theta\bar{\mathbb{M}}^{1-\nu}] \hat{c}_t + \theta\bar{\mathbb{M}}^{1-\nu}(\hat{m}_t - \hat{p}_t).$$

$$\Rightarrow \bar{X}^{1-\nu}(\hat{c}_t - \hat{x}_t) = \theta\bar{\mathbb{M}}^{1-\nu} [\hat{c}_t - (\hat{m}_t - \hat{p}_t)].$$

$$\Rightarrow \hat{c}_t - \hat{x}_t = \underbrace{\theta \left( \frac{\bar{\mathbb{M}}}{\bar{X}} \right)^{1-\nu}}_{\chi} [\hat{c}_t - (\hat{m}_t - \hat{p}_t)]. \quad (***)$$

$$\text{Define } \left( \frac{\bar{\mathbb{M}}}{\bar{P}} \right) / \bar{C} = k_m$$

$k_m$  : inverse consumption velocity, low  $k_m$  high velocity

$$\begin{aligned} \text{Let } \chi &= \theta \frac{\left( \frac{\bar{\mathbb{M}}}{\bar{P}} \right)^{1-\nu}}{\bar{X}^{1-\nu}} = \theta \frac{\left( \frac{\bar{\mathbb{M}}}{\bar{P}} \right)^{1-\nu}}{(1-\theta)\bar{C}^{1-\nu} + \theta \left( \frac{\bar{\mathbb{M}}}{\bar{P}} \right)^{1-\nu}} = \frac{\theta \left( \left( \frac{\bar{\mathbb{M}}}{\bar{P}} \right) / \bar{C} \right)^{1-\nu}}{1 - \theta + \theta \left( \left( \frac{\bar{\mathbb{M}}}{\bar{P}} \right) / \bar{C} \right)^{1-\nu}} \\ &= \frac{\theta k_m^{1-\nu}}{1 - \theta + \theta k_m^{1-\nu}} \end{aligned}$$

$\chi$  : steady-state expenditure share of money inside the CES aggregator

$$\begin{aligned} \bar{\mathbb{M}}_t &= C_t(1 - e^{-it})^{-1/\nu} \left( \frac{\theta}{1-\theta} \right)^{1/\nu} \Rightarrow \left( \frac{\bar{\mathbb{M}}}{\bar{P}} \right) / \bar{C} = (1-\beta)^{-1/\nu} \left( \frac{\theta}{1-\theta} \right)^{1/\nu} \\ &\Rightarrow k_m = \left[ \frac{\theta}{(1-\beta)(1-\theta)} \right]^{1/\nu} \end{aligned}$$

$k_m$  increases with money weight  $\theta$ ,  $\beta \uparrow \Rightarrow$  opportunity cost of holding money  $\downarrow$

Plug (\*\*\*) into (\*\*), we have

$$\hat{w}_t - \hat{p}_t = \sigma \hat{c}_t + \varphi \hat{n}_t + (\nu - \sigma)\chi [\hat{c}_t - (\hat{m}_t - \hat{p}_t)]$$

Plug (\*) into the above equation:

$$\hat{w}_t - \hat{p}_t = \sigma \hat{c}_t + \varphi \hat{n}_t + (\nu - \sigma) \chi \left[ \hat{c}_t - \hat{y}_t + \eta \hat{i}_t \right]$$

$$\Rightarrow \quad \hat{w}_t - \hat{p}_t = \sigma \hat{c}_t + \varphi \hat{n}_t + \omega \hat{i}_t$$

$$\text{where} \quad \omega = (\nu - \sigma) \chi \eta$$

$\omega$  : like nominal interest rate and labor supply

$$\chi = \frac{\theta^{1/\nu} (1 - \beta)^{1 - \frac{1}{\nu}}}{(1 - \theta)^{1/\nu} + \theta^{1/\nu} (1 - \beta)^{1 - \frac{1}{\nu}}} = \frac{\left(\frac{\theta}{1 - \theta}\right)^{1/\nu} (1 - \beta)^{1 - \frac{1}{\nu}}}{1 + \left(\frac{\theta}{1 - \theta}\right)^{1/\nu} (1 - \beta)^{1 - \frac{1}{\nu}}}$$

$$\Rightarrow \quad \chi = \frac{k_m (1 - \beta)}{1 + k_m (1 - \beta)}$$

$$k_m (1 - \beta) = \left(\frac{\theta}{1 - \theta}\right)^{1/\nu} (1 - \beta)^{1 - \frac{1}{\nu}}$$

$$\Rightarrow \quad \omega = \chi \eta (\nu - \sigma) = \frac{k_m (1 - \beta)}{1 + k_m (1 - \beta)} \cdot \frac{1}{\nu (e^{\bar{i}} - 1)} \cdot (\nu - \sigma)$$

$$e^{\bar{i}} = \beta^{-1} \Rightarrow e^{\bar{i}} - 1 = \frac{1 - \beta}{\beta}$$

$$\Rightarrow \quad \omega = \frac{k_m \beta (\nu - \sigma)}{\nu [1 + k_m (1 - \beta)]}$$

$$\text{sign}(\omega) = \text{sign}\left(1 - \frac{\sigma}{\nu}\right) \Rightarrow \quad \omega > 0 \iff \nu > \sigma$$

$$\eta = \frac{1}{\nu (e^{\bar{i}} - 1)}$$

$$\text{for small } \bar{i}, \quad e^{\bar{i}} - 1 \approx \bar{i} \Rightarrow \eta \approx \frac{1}{\nu \bar{i}} \Rightarrow \nu \approx \frac{1}{\eta \bar{i}}$$

$$\text{since } \bar{i} \text{ is small, } \nu = \frac{1}{\eta \bar{i}} \gg \sigma \Rightarrow \omega > 0$$

$$\begin{cases} \hat{w}_t - \hat{p}_t = \sigma \hat{c}_t + \varphi \hat{n}_t + \omega \hat{i}_t, \\ \hat{m}_t - \hat{p}_t = \hat{y}_t - \eta \hat{i}_t, \end{cases} \quad \hat{c}_t - \hat{x}_t = \chi [\hat{c}_t - (\hat{m}_t - \hat{p}_t)]$$

$$\hat{i}_t \uparrow, \eta > 0 \Rightarrow \hat{m}_t - \hat{p}_t \downarrow \Rightarrow \hat{x}_t \downarrow \Rightarrow U_{C,t} \downarrow \Rightarrow U_{\text{CM}} > 0$$

Money and consumption are complements.

$$\hat{i}_t \uparrow, \eta > 0 \Rightarrow \hat{n}_t \downarrow \quad \text{given real wage and consumption}$$

Money and labor move in the same direction: Less liquidity,  $U_{C,t} \downarrow$ , consumption is less valuable, and the benefit of working is lower.

**Log-linearize:**

$$Q_t = \beta E_t \left[ \frac{U_{C,t+1}}{U_{C,t}} \frac{P_t}{P_{t+1}} \right] = \beta E_t \left[ \left( \frac{C_{t+1}}{C_t} \right)^{-\nu} \left( \frac{X_{t+1}}{X_t} \right)^{\nu-\sigma} \frac{P_t}{P_{t+1}} \right]$$

$$Q_t = e^{-i_t}, \quad \beta = e^{-\rho}, \quad \Pi_{t+1} = \frac{P_{t+1}}{P_t}, \quad \pi_{t+1} = \log \Pi_{t+1} = \log P_{t+1} - \log P_t$$

$$\Rightarrow e^{-i_t} = e^{-\rho} E_t \left[ e^{-\nu(\log C_{t+1} - \log C_t) + (\nu-\sigma)(\log X_{t+1} - \log X_t) - \pi_{t+1}} \right]$$

$$\Rightarrow \begin{cases} -i_t = -\rho + E_t[-\nu(\log C_{t+1} - \log C_t) + (\nu-\sigma)(\log X_{t+1} - \log X_t) - \pi_{t+1}], \\ -\bar{i} = -\rho + E_t[-\nu(\log \bar{C} - \log \bar{C}) + (\nu-\sigma)(\log \bar{X} - \log \bar{X}) - \bar{\pi}] \end{cases}$$

$$\bar{\pi} = 0$$

$$\Rightarrow \hat{i}_t = i_t - \bar{i} = E_t[\nu(\hat{c}_{t+1} - \hat{c}_t) - (\nu-\sigma)(\hat{x}_{t+1} - \hat{x}_t) + \pi_{t+1}]$$

$$\Rightarrow \hat{i}_t - E_t \pi_{t+1} = \nu E_t(\hat{c}_{t+1} - \hat{c}_t) - (\nu-\sigma) E_t(\hat{x}_{t+1} - \hat{x}_t)$$

$$\text{since} \quad \hat{c}_t - \hat{x}_t = \chi [\hat{c}_t - (\hat{m}_t - \hat{p}_t)]$$

$$\hat{m}_t - \hat{p}_t = \hat{c}_t - \eta \hat{i}_t \quad \Rightarrow \quad \hat{c}_t - \hat{x}_t = \chi \eta \hat{i}_t \quad \Rightarrow \quad \hat{x}_t = \hat{c}_t - \chi \eta \hat{i}_t$$

$$\Rightarrow \hat{i}_t - E_t \pi_{t+1} = \nu E_t \Delta \hat{c}_{t+1} - (\nu-\sigma) E_t \Delta \hat{c}_{t+1} + (\nu-\sigma) \chi \eta E_t \Delta \hat{i}_{t+1}$$

$$\Rightarrow \hat{i}_t - E_t \pi_{t+1} = \sigma E_t \hat{c}_{t+1} - \sigma \hat{c}_t + \omega E_t \Delta \hat{i}_{t+1}$$

$$\Rightarrow \sigma E_t \Delta \hat{c}_{t+1} = \hat{i}_t - E_t \pi_{t+1} - \omega E_t \Delta \hat{i}_{t+1}$$

$$\Rightarrow \sigma E_t \hat{c}_{t+1} - \sigma \hat{c}_t = \hat{i}_t - E_t \pi_{t+1} - \omega E_t \Delta \hat{i}_{t+1}$$

$$\Rightarrow \hat{c}_t = E_t \hat{c}_{t+1} - \frac{1}{\sigma} [\hat{i}_t - E_t \pi_{t+1} - \omega E_t \Delta \hat{i}_{t+1}] \quad \dots (*)$$

$$\text{where} \quad \omega = (\nu-\sigma) \chi \eta, \quad \chi = \frac{k_m(1-\beta)}{1+k_m(1-\beta)}, \quad k_m = \frac{\bar{M}/\bar{P}}{\bar{C}}, \quad \eta = \frac{1}{\nu(e^{\bar{i}} - 1)}.$$

In (\*) we apply market clearing  $\hat{c}_t = \hat{y}_t$

households smooth consumption  $\rightarrow$  how responsive current spending is to intertemporal incentive

$$\Rightarrow \hat{y}_t = E_t \hat{y}_{t+1} - \frac{1}{\sigma} [\hat{i}_t - E_t \pi_{t+1} - \omega E_t \Delta \hat{i}_{t+1}]$$

current demand/output

= expected future output/demand – IES  $\times$  expected real interest rate – forward guidance

For  $\nu > \sigma \Rightarrow \omega > 0$ . When  $E_t \Delta \hat{i}_{t+1} \uparrow$ ,  $\hat{c}_t \uparrow$ . forward guidance

If  $E_t \Delta \hat{i}_{t+1} \uparrow$ , then future money holdings become less attractive, future liquidity services decline, and households shift some consumption from the future to today; hence  $\hat{c}_t \uparrow$ .

$$\frac{W_t}{P_t} = (1 - \alpha)A_t N_t^{-\alpha}, \quad Y_t = A_t N_t^{1-\alpha}$$

$$\hat{w}_t - \hat{p}_t = \hat{a}_t - \alpha \hat{n}_t, \quad \hat{y}_t = \hat{a}_t + (1 - \alpha) \hat{n}_t$$

$$\Rightarrow \hat{n}_t = \frac{1}{1 - \alpha} (\hat{y}_t - \hat{a}_t)$$

$$\Rightarrow \hat{w}_t - \hat{p}_t = \hat{y}_t - \hat{n}_t$$

$$\hat{w}_t - \hat{p}_t = \sigma \hat{c}_t + \varphi \hat{n}_t + \omega \hat{i}_t$$

$$\Rightarrow \hat{y}_t - \hat{n}_t = \sigma \hat{y}_t + \varphi \hat{n}_t + \omega \hat{i}_t$$

$$\Rightarrow (1 - \sigma) \hat{y}_t - (1 + \varphi) \hat{n}_t = \omega \hat{i}_t$$

$$\Rightarrow \hat{n}_t = \frac{1 - \sigma}{1 + \varphi} \hat{y}_t - \frac{\omega}{1 + \varphi} \hat{i}_t$$

$$\hat{y}_t = \hat{a}_t + (1 - \alpha) \hat{n}_t$$

$$\Rightarrow \hat{y}_t = \hat{a}_t + (1 - \alpha) \left[ \frac{1 - \sigma}{1 + \varphi} \hat{y}_t - \frac{\omega}{1 + \varphi} \hat{i}_t \right]$$

$$\Rightarrow \hat{y}_t = \frac{1 + \varphi}{\alpha(1 - \sigma) + \sigma + \varphi} \hat{a}_t - \frac{(1 - \alpha)\omega}{\alpha(1 - \sigma) + \sigma + \varphi} \hat{i}_t \quad (1)$$

$$\Rightarrow \hat{y}_t - \hat{n}_t = \sigma \hat{y}_t + \varphi \hat{n}_t + \omega \hat{i}_t$$

$$\Rightarrow (1 - \sigma) \hat{y}_t - (1 + \varphi) \hat{n}_t = \omega \hat{i}_t$$

$$\Rightarrow \hat{n}_t = \frac{1 - \sigma}{1 + \varphi} \hat{y}_t - \frac{\omega}{1 + \varphi} \hat{i}_t$$

$$\hat{y}_t = \hat{a}_t + (1 - \alpha) \hat{n}_t$$

$$\Rightarrow \hat{y}_t = \hat{a}_t + (1 - \alpha) \left[ \frac{1 - \sigma}{1 + \varphi} \hat{y}_t - \frac{\omega}{1 + \varphi} \hat{i}_t \right]$$

$$\Rightarrow \hat{y}_t = \frac{1 + \varphi}{\alpha(1 - \sigma) + \sigma + \varphi} \hat{a}_t - \frac{(1 - \alpha)\omega}{\alpha(1 - \sigma) + \sigma + \varphi} \hat{i}_t \quad (1)$$

$\hat{y}_t$  : interest rate  $\rightarrow$  real activity  $\rightarrow$  transmission coefficient

comes from money-in-the-utility non-separability.

Money is non-neutral (interest rate / monetary policy affects real output)

But the function alone cannot pin down it; we need a monetary policy rule.

Write the system of equations in terms of  $(\hat{y}_t, \pi_t, \hat{i}_t)$ .

$$\text{Euler equation: } \hat{y}_t = E_t \hat{y}_{t+1} - \frac{1}{\sigma} [\hat{i}_t - E_t \pi_{t+1} - \omega E_t \Delta \hat{i}_{t+1}] \quad (2)$$

Also known as IS equation

$$\text{Monetary policy rule: } \hat{i}_t = \phi_\pi \pi_t + v_t \quad (3)$$

$$\text{where } v_t = \rho_v v_{t-1} + \varepsilon_t^v \quad (4)$$

$$\hat{a}_t = \rho_a \hat{a}_{t-1} + \varepsilon_t^a \quad (5)$$

Combine (1)–(5), use undetermined coefficients:

$$\begin{aligned} \pi_t &= A_\pi \hat{a}_t + B_\pi v_t, & \hat{i}_t &= A_i \hat{a}_t + B_i v_t, & \hat{y}_t &= A_y \hat{a}_t + B_y v_t. \\ \Rightarrow \left\{ \begin{aligned} \pi_t &= -\frac{\sigma(1-\rho_a)\psi_{ya}}{\phi_\pi(1+\omega\psi)(1-\theta\rho_a)} \hat{a}_t - \frac{1+(1-\rho_v)\omega\psi}{\phi_\pi(1+\omega\psi)(1-\theta\rho_v)} v_t, \\ \hat{i}_t &= -\frac{\sigma(1-\rho_a)\psi_{ya}}{(1+\omega\psi)(1-\theta\rho_a)} \hat{a}_t - \frac{\rho_v}{\phi_\pi(1+\omega\psi)(1-\theta\rho_v)} v_t, \\ \hat{y}_t &= \psi_{ya} \left( 1 + \frac{\sigma(1-\rho_a)\psi_{yi}}{(1+\omega\psi)(1-\theta\rho_a)} \right) \hat{a}_t + \frac{\rho_v\psi_{yi}}{\phi_\pi(1+\omega\psi)(1-\theta\rho_v)} v_t. \end{aligned} \right. \\ \text{where } \theta &= \frac{1+\omega\phi\phi_\pi}{(1+\omega\phi)\phi_\pi}, & \psi &= \frac{\alpha+\phi}{\sigma(1-\alpha)+\alpha+\phi}. \end{aligned}$$

Intuition:  $\psi$  is a summary slope coming from the real-side block, capturing how strongly real activities / labor market tightness maps into wages and output in the log-linear world.

$\theta$  is a dynamic feedback object mixing the Taylor coefficient  $\phi_\pi$ , the forward-guidance strength  $\omega$ , and the real-side slope.

When  $\omega = 0$ ,  $\theta$  collapses to  $1/\phi_\pi$ , which is the usual policy-feedback scale.

When  $\omega \neq 0$ ,  $\theta$  changes because expected changes in it enter the IS equation.

## Impact of the monetary policy

Conditional on an exogenous MP shock  $\varepsilon_t^v$  to  $v_t$  :  $\hat{y}_t = \psi_{ya}\hat{a}_t - \psi_{yi}\hat{i}_t \Rightarrow \frac{d\hat{y}_t}{d\hat{i}_t} = -\psi_{yi}$ .

$$\text{Assume } \sigma = \phi = 1, \quad \alpha = \frac{1}{3} \quad \Rightarrow \quad \psi_{yi} = \frac{\omega}{3}.$$

$$\omega = \frac{k_m \beta (1 - \frac{\sigma}{\nu})}{1 + k_m(1 - \beta)} \approx k_m \quad \text{with } \nu \text{ large and } \beta \approx 1.$$

$$\Rightarrow \quad \psi_{yi} \approx \frac{k_m}{3}.$$

$$\text{Recall } k_m = \frac{\bar{M}/\bar{P}}{\bar{C}},$$

how much liquidity stock households want to hold per unit of spending (cash buffer relative to spending)

Using M1, the money with highest liquidity:  $k_m \approx 0.3 \Rightarrow \psi_{yi} \approx 0.1$ .

If  $i_t$  increases by 1% annually (0.25% quarterly),  $y_t$  decreases by 0.025% quarterly.

Using M2, the money with highest liquidity + 2nd highest liquidity:  $k_m \approx 3 \Rightarrow \psi_{yi} \approx 1$ .

If  $i_t$  increases by 1% annually,  $y_t$  decreases by 1% annually.

**\* If money is quantitatively large, MIU channel becomes stronger.**

## Deficiency of MIU model

$$\frac{d\pi_t}{d\hat{i}_t} = \frac{d\pi_t/dv_t}{d\hat{i}_t/dv_t} = (1 + (1 - \rho_v)\omega\psi)\rho_v^{-1} > 0,$$

$$\frac{d\hat{r}_t}{d\hat{i}_t} = 1 - \frac{dE_t(\pi_{t+1})/dv_t}{d\hat{i}_t/dv_t} = -(1 - \rho_v)\omega\psi < 0.$$

A monetary policy shock that raises the nominal interest rate and lowers output also results in:

- (i) inflation tending to increase;
- (ii) the real interest rate tending to decrease

$$(r_t = i_t - E_t\pi_{t+1}).$$

This contradicts the data: contractionary monetary policy shocks raise real interest rates and lower inflation.

In the data, money is neutral in the long run once prices adjust, but non-neutral in the short run. In the model, long-run and short-run effects are the same.

## Uniqueness of the nominal system

In a flexible-price (classical) economy, the real side pins down the real interest rate  $\hat{r}_t$ . The question is: what must monetary policy do to ensure that inflation  $\pi_t$  — and hence the price level  $p_t$  — is uniquely determined?

The Fisher relation is

$$\hat{i}_t = E_t[\pi_{t+1}] + \hat{r}_t, \quad \Rightarrow \quad E_t[\pi_{t+1}] = \hat{i}_t - \hat{r}_t.$$

Since  $\hat{r}_t$  is pinned down by the real economy, if the central bank exogenously sets  $\hat{i}_t$  as some stationary process, or simply sets  $\hat{i}_t = 0$ , then the Fisher equation pins down expected inflation only, not realized inflation.

Therefore, monetary policy must do more than fix the nominal interest rate exogenously: it must provide a rule that anchors the nominal system and uniquely determines inflation.

$$\Rightarrow \quad E_t[\pi_{t+1}] = -\hat{r}_t \quad \Rightarrow \quad \pi_{t+1} = -\hat{r}_t + \zeta_{t+1}, \quad E_t[\zeta_{t+1}] = 0.$$

Any zero-mean innovation can be added without violating Fisher equation  $\Rightarrow$  indeterminacy.

With an exogenous  $\hat{i}_t$ , there are infinitely many inflation paths consistent with equilibrium.

$$\Rightarrow \quad \text{sunspot driven (non-fundamental factors).}$$

Since  $\hat{p}_t = \pi_t + \hat{p}_{t-1}$ ,  $\pi_t$  is indetermined  $\Rightarrow \hat{p}_t$  is also indetermined (not unique).

Other nominal variables are also indetermined:

$$\hat{m}_t = \hat{p}_t + \hat{y}_t - \eta \hat{i}_t \quad \Rightarrow \quad \hat{m}_t \text{ indetermined} \quad (\text{nominal money})$$

$$\hat{w}_t = \hat{w}_t^r + \hat{p}_t \quad \Rightarrow \quad \hat{W}_t \text{ indetermined} \quad (\text{nominal wage})$$

At steady state,

$$P_t = \bar{P} \Rightarrow \bar{\pi} = 0 \Rightarrow \bar{i} = \bar{r} = \rho,$$

so that

$$\hat{r}_t = r_t - \rho.$$

A simple inflation-based interest rate rule (Taylor style) is

$$\hat{i}_t = \phi_\pi \pi_t, \quad \phi_\pi \geq 0.$$

$$\hat{i}_t = E_t[\pi_{t+1}] + \hat{r}_t,$$

we obtain

$$\phi_\pi \pi_t = E_t[\pi_{t+1}] + \hat{r}_t. \quad (*)$$

This is a forward-looking difference equation in inflation.

**Case 1:**  $\phi_\pi > 1$

Iterate forward:

$$\pi_t = \frac{1}{\phi_\pi} E_t[\pi_{t+1}] + \frac{1}{\phi_\pi} \hat{r}_t \quad \Rightarrow \quad \pi_t = \sum_{k=0}^{\infty} \phi_\pi^{-(k+1)} E_t[\hat{r}_{t+k}],$$

where the forward sum converges and yields a unique bounded solution.

Bond Euler:

$$\hat{i}_t - E_t[\pi_{t+1}] = \sigma E_t \Delta \hat{c}_{t+1}, \quad \hat{i}_t - E_t[\pi_{t+1}] = \hat{r}_t,$$

so

$$\hat{r}_t = \sigma E_t \Delta \hat{c}_{t+1}.$$

Since  $\Delta \hat{c}_{t+1} = \Delta \hat{y}_{t+1}$ , and

$$\hat{y}_t = \psi_{ya} \hat{a}_t \quad \Rightarrow \quad \Delta \hat{y}_{t+1} = \psi_{ya} \Delta \hat{a}_{t+1},$$

we obtain

$$\hat{r}_t = \sigma \psi_{ya} E_t \Delta \hat{a}_{t+1}.$$

If productivity follows

$$\hat{a}_{t+1} = \rho_a \hat{a}_t + \varepsilon_{t+1}^a,$$

then

$$E_t[\hat{a}_{t+1}] = \rho_a \hat{a}_t, \quad E_t[\Delta \hat{a}_{t+1}] = E_t[\hat{a}_{t+1} - \hat{a}_t] = (\rho_a - 1) \hat{a}_t.$$

Hence,

$$\hat{r}_t = \sigma \psi_{ya} (\rho_a - 1) \hat{a}_t.$$

Guess a linear solution:

$$\pi_t = K \hat{a}_t, \quad E_t \hat{a}_{t+1} = \rho_a \hat{a}_t.$$

Substituting into

$$\phi_\pi \pi_t = E_t \pi_{t+1} + \hat{r}_t$$

gives

$$\phi_\pi K \hat{a}_t = K \rho_a \hat{a}_t + \hat{r}_t = K \rho_a \hat{a}_t - \sigma \psi_{ya} (1 - \rho_a) \hat{a}_t,$$

so

$$\phi_\pi K - K \rho_a = -\sigma \psi_{ya} (1 - \rho_a),$$

and therefore

$$K = -\frac{\sigma \psi_{ya} (1 - \rho_a)}{\phi_\pi - \rho_a}.$$

Thus,

$$\pi_t = -\frac{\sigma \psi_{ya} (1 - \rho_a)}{\phi_\pi - \rho_a} \hat{a}_t.$$

The larger the  $\phi_\pi$ , the smaller the variation of inflation caused by  $\hat{a}_t$ . Central bank raises it more when inflation rises. Any shock that pushes up  $\pi_t$  implies a larger rise in  $i_t$ ; through the Fisher equation, expectations are forced to be consistent only with the saddle path.

**Case 2:**  $\phi_\pi \leq 1$  Forward iteration breaks. Since  $\phi_\pi^{-1} \geq 1$ , boundedness no longer pins down a unique  $\pi_t$ .

To satisfy the Fisher equation,

$$\phi_\pi \pi_t = E_t[\pi_{t+1}] + \hat{r}_t.$$

The stable solution is

$$\pi_{t+1} = \phi_\pi \pi_t - \hat{r}_t + \xi_{t+1}, \quad E_t[\xi_{t+1}] = 0,$$

with sunspot shocks. Inflation and the price level are indetermined.

If the central bank responds less than one-for-one to inflation (weak response), expected inflation can move around without being forcefully pushed back by policy. This leaves room for self-fulfilling beliefs about inflation (sunspot shocks), just as in the exogenous  $i_t$  case.

Determinacy requires the Taylor principle: raise the nominal rate more than one-for-one with inflation, that is,  $\phi_\pi > 1$ .

The system is

$$E_t \pi_{t+1} = \phi_\pi \pi_t - \hat{r}_t.$$

Inflation  $\pi_t$  is a jump variable. By the Blanchard–Kahn condition, to pin down a unique equilibrium we need

$$\# \text{ unstable roots} = \# \text{ jump variables}.$$

From

$$E_t \pi_{t+1} = \phi_\pi \pi_t,$$

the eigenvalue is  $\phi_\pi$ . If  $\phi_\pi > 1$ , there is an unstable root, so the BK condition is satisfied. If  $\phi_\pi \leq 1$ , there is no unstable root to discipline the jump variable, and we get many bounded paths: sunspot indeterminacy.

**Optimal Monetary Policy** Assume a social planner takes the role of the central banker. Social planner's problem:

$$\max U\left(C_t, \frac{M_t}{P_t}, N_t\right) \quad \text{s.t.} \quad C_t = A_t N_t^{1-\alpha}.$$

The Lagrangian is

$$\mathcal{L} = U\left(C_t, \frac{M_t}{P_t}, N_t\right) + \lambda_t (A_t N_t^{1-\alpha} - C_t).$$

F.O.C.:

$$-\frac{U_{N,t}}{U_{C,t}} = (1-\alpha)A_t N_t^{-\alpha} = W_t = MPL_t = MRS_{N,C},$$

and

$$U_{M,t} = 0.$$

In competitive equilibrium,

$$\frac{U_{M,t}}{U_{C,t}} = 1 - e^{-i_t} \approx i_t.$$

Social and individual optima coincide iff  $i_t = 0$  for all  $t$ , which is the Friedman rule.

In the long run, with the Fisher equation,

$$i = \pi + r = 0 \quad \Rightarrow \quad \pi = -r = -\rho < 0,$$

which is a moderate deflation.

In the short run,  $i_t = 0$  does not guarantee determinacy. Instead,

$$E_t \pi_{t+1} = -r_t,$$

or equivalently,

$$\pi_{t+1} = -r_t + \xi_{t+1}, \quad E_t[\xi_{t+1}] = 0,$$

so any sunspot shock can be added.

Even though money balances and other nominal terms are indetermined, real variables are pinned down by

$$MRS = MPL.$$

Thus indeterminacy does not have welfare consequences.

To avoid indeterminacy, let

$$i_t = \phi (r_{t-1} + \pi_t), \quad \phi > 1.$$

Then

$$i_t = E_t \pi_{t+1} + r_t$$

$$E_t[i_{t+s}] = \phi^s i_t.$$

Since  $\phi > 1$ , as  $s \rightarrow \infty$ ,  $i_t$  has to approach 0.

$$\Rightarrow \quad \pi_t = -r_{t-1}, \text{ so inflation is fully determined.}$$

Any policy setting that guarantees that current inflation moves inversely, and one-for-one, with  $r_{t-1}$  will imply  $i_t = 0$  and an efficient amount of real balances.

### Supplementary: IS equation under the separable case in the MIU model

$$U(C_t, \mathbb{M}_t, N_t) = \frac{C_t^{1-\sigma}}{1-\sigma} + \frac{(\mathbb{M}_t)^{1-\nu}}{1-\nu} - \frac{N_t^{1+\varphi}}{1+\varphi}.$$

From the bond Euler F.O.C.,

$$Q_t = \beta E_t \left[ \frac{U_{C,t+1}}{U_{C,t}} \frac{P_t}{P_{t+1}} \right].$$

Since

$$U_{C,t} = C_t^{-\sigma},$$

we have

$$Q_t = \beta E_t \left[ \left( \frac{C_{t+1}}{C_t} \right)^{-\sigma} \frac{P_t}{P_{t+1}} \right].$$

Define

$$i_t = -\log Q_t, \quad \rho = -\log \beta, \quad \pi_{t+1} = \log P_{t+1} - \log P_t.$$

Then

$$-i_t = -\rho + E_t[-\sigma(\log C_{t+1} - \log C_t) - \pi_{t+1}].$$

At steady state,

$$-\bar{i} = -\rho + E_t[-\sigma(\log \bar{C} - \log \bar{C}) - \bar{\pi}],$$

with

$$\bar{\pi} = 0.$$

Hence

$$\bar{i} = \rho.$$

Subtracting the steady state and using hats,

$$\sigma E_t(\hat{c}_{t+1} - \hat{c}_t) = \hat{i}_t - E_t\pi_{t+1}.$$

Thus

$$\hat{c}_t = E_t\hat{c}_{t+1} - \frac{1}{\sigma} [\hat{i}_t - E_t\pi_{t+1}].$$

Since

$$\hat{y}_t = \hat{c}_t,$$

we get

$$\hat{y}_t = E_t\hat{y}_{t+1} - \frac{1}{\sigma} [\hat{i}_t - E_t\pi_{t+1}].$$

Compare to the classical monetary model: the same.

Compare to the inseparable case:  $\nu = \sigma$ , and the term  $-\omega E_t\Delta\hat{i}_{t+1}$  disappears?

**Supplement: Interpret**

$$\hat{y}_t = E_t\hat{y}_{t+1} - \frac{1}{\sigma} [\hat{i}_t - E_t\pi_{t+1} - \omega E_t\Delta\hat{i}_{t+1}].$$

1. The term

$$i_t - E_t\pi_{t+1}$$

is the opportunity cost of consuming today instead of tomorrow.

Suppose you have 1 RMB.

If you buy goods today, you get

$$\frac{1}{P_t}$$

units of goods.

If you save to tomorrow and then buy goods, you get

$$\frac{1 + i_t}{P_{t+1}}$$

units of goods. Therefore,

$$E_t \left[ \frac{(1 + i_t)/P_{t+1}}{1/P_t} \right] = E_t \left[ \frac{1 + i_t}{P_{t+1}/P_t} \right] = \frac{1 + i_t}{1 + E_t\pi_{t+1}} \approx i_t - E_t\pi_{t+1}.$$

So if

$$\hat{i}_t - E_t\pi_{t+1} \uparrow,$$

then

$$\hat{y}_t \downarrow,$$

because the opportunity cost of consuming today is higher.

2. The term

$$\omega E_t \hat{i}_{t+1}$$

is the forward-guidance term.

If tomorrow's nominal rate  $i_{t+1}$  is expected to be high, then holding money tomorrow is expensive. Households therefore want to hold fewer real balances at  $t + 1$ :

$$\frac{M_{t+1}}{P_{t+1}} \downarrow.$$

With  $\omega > 0$ , there is within-period substitution between  $C_t$  and  $\frac{M_{t+1}}{P_{t+1}}$ , in addition to the usual intertemporal substitution of  $X_t$ .

Holding more real balances,

$$\frac{M_{t+1}}{P_{t+1}} \uparrow,$$

increases the marginal utility of consumption,  $MU_{C,t+1}$ , so future consumption becomes more valuable. Conversely,

$$\frac{M_{t+1}}{P_{t+1}} \downarrow$$

makes future consumption less valuable and induces households to consume more today.

In particular,

$$U_{C,t} = (1 - \theta) X_t^{\nu - \sigma} C_t^{-\nu}, \quad X_t \propto \mathbb{M}_t \quad \Rightarrow \quad U_{CM} > 0.$$

Consumption is more enjoyable with more liquidity.

**Supplement for computation: Log-linearized bond Euler equation** Define

$$i_t = -\log Q_t \Rightarrow e^{-i_t} = Q_t, \quad \rho = -\log \beta \Rightarrow e^{-\rho} = \beta, \quad \pi_{t+1} = \log P_{t+1} - \log P_t,$$

and

$$r_t = i_t - E_t[\pi_{t+1}].$$

Define hats as level deviations:

$$\hat{i}_t = i_t - \bar{i}, \quad \hat{\pi}_{t+1} = \pi_{t+1} - \bar{\pi}, \quad \bar{\pi} = 0 \Rightarrow \hat{\pi}_{t+1} = \pi_{t+1}.$$

Also,

$$r_t - \bar{r} = i_t - \bar{i} - E_t[\pi_{t+1} - \bar{\pi}] \Rightarrow \hat{r}_t = \hat{i}_t - E_t[\hat{\pi}_{t+1}].$$

From the bond Euler equation,

$$Q_t = \beta E_t \left[ \left( \frac{C_{t+1}}{C_t} \right)^{-\sigma} \frac{P_t}{P_{t+1}} \right] = \beta E_t \left[ e^{-\sigma(\log C_{t+1} - \log C_t) - \pi_{t+1}} \right],$$

so

$$e^{-i_t} = e^{-\rho} E_t \left[ e^{-\sigma(\log C_{t+1} - \log C_t) - \pi_{t+1}} \right].$$

Let

$$\hat{c}_{t+1} = \log C_{t+1} - \log \bar{C}, \quad \hat{c}_t = \log C_t - \log \bar{C} \Rightarrow \log C_{t+1} - \log C_t = \hat{c}_{t+1} - \hat{c}_t.$$

Then

$$e^{-i_t} = e^{-\rho} E_t \left[ e^{-\sigma(\hat{c}_{t+1} - \hat{c}_t) - \pi_{t+1}} \right].$$

Taking logs,

$$-i_t = -\rho + E_t[-\sigma(\hat{c}_{t+1} - \hat{c}_t) - \pi_{t+1}].$$

At steady state,

$$-\bar{i} = -\rho + E_t[0 - 0] \Rightarrow \bar{i} = \rho.$$

Hence

$$-(i_t - \bar{i}) = E_t[-\sigma(\hat{c}_{t+1} - \hat{c}_t)] - E_t[\pi_{t+1} - \bar{\pi}],$$

or equivalently,

$$\hat{i}_t = \sigma E_t[\Delta \hat{c}_{t+1}] + E_t[\hat{\pi}_{t+1}], \quad \Delta \hat{c}_{t+1} = \hat{c}_{t+1} - \hat{c}_t.$$

Therefore,

$$\sigma E_t[\Delta \hat{c}_{t+1}] = \hat{i}_t - E_t[\hat{\pi}_{t+1}] = \hat{r}_t.$$

Using  $\hat{c}_t = \hat{y}_t$ , we obtain

$$\sigma E_t[\Delta \hat{y}_{t+1}] = \hat{r}_t.$$

Since

$$E_t[\Delta \hat{y}_{t+1}] = E_t \hat{y}_{t+1} - \hat{y}_t, \quad \hat{y}_t = \psi_{ya} \hat{a}_t \Rightarrow E_t[\Delta \hat{y}_{t+1}] = \psi_{ya} E_t[\Delta \hat{a}_{t+1}],$$

it follows that

$$\hat{r}_t = \sigma \psi_{ya} E_t[\Delta \hat{a}_{t+1}].$$

Let

$$\log A_{t+1} = (1 - \psi) \log \bar{A} + \varphi \log A_t + \varepsilon_{t+1}, \quad \bar{A} = 1.$$

Then

$$\log \bar{A} = \psi \log \bar{A}, \quad \hat{a}_{t+1} = \psi \hat{a}_t + \varepsilon_{t+1}.$$

Taking expectations,

$$E_t[\hat{a}_{t+1}] = \psi \hat{a}_t + E_t[\varepsilon_{t+1}], \quad \varepsilon_t \sim \text{i.i.d. } N(0, \sigma^2) \Rightarrow E_t[\varepsilon_{t+1}] = 0,$$

so

$$E_t[\hat{a}_{t+1}] = \psi \hat{a}_t, \quad E_t[\hat{a}_{t+1} - \hat{a}_t] = (\psi - 1) \hat{a}_t.$$

Let

$$\Delta \hat{a}_{t+1} = \hat{a}_{t+1} - \hat{a}_t \Rightarrow E_t[\Delta \hat{a}_{t+1}] = (\psi - 1) \hat{a}_t.$$

Plugging this into  $\hat{r}_t = \sigma \psi_{ya} E_t[\Delta \hat{a}_{t+1}]$ , we get

$$\hat{r}_t = \sigma \psi_{ya} E_t[\Delta \hat{a}_{t+1}] = \sigma \psi_{ya} (\psi - 1) \hat{a}_t.$$

Since  $\psi \in (0, 1)$  and  $\psi_{ya} > 0$ , the coefficient on  $\hat{a}_t$  is negative.